

RUSSELLVILLE 4 SSE, AL, USA

WMO: 750060

Lat: 34.4535N

Lon: 87.7104W

Elev: 220

StdP: 98.72

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 33.7	(d) N/A	(e) 32.6	(f) N/A	(g) 31.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(d) DB	(e) -12.9	(f) 35.4	(g) 3.0	(h) 2.0	(i) -15.1	(j) 36.9	(k) -16.8	(l) 38.1	(m) -18.4	(n) 39.2	(o) -20.6	(p) 40.7	(q) N/A
(a) N/A	(b) N/A	(c) N/A	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.4	6.9	11.3	15.7	20.0	24.2	25.3	25.0	22.0	15.8	9.6	7.0
	DBStd	8.62	6.34	6.10	5.58	4.38	3.71	2.24	2.10	2.22	3.40	4.81	5.07	5.63
	HDD10.0	582	197	121	53	7	0	0	0	0	0	7	67	129
	HDD18.3	1841	433	321	223	102	29	1	0	0	10	107	264	352
	CDD10.0	2647	23	36	95	176	311	425	474	466	361	188	56	37
	CDD18.3	863	0	2	6	22	82	176	216	207	121	29	3	1
	CDH23.3	7861	0	3	53	199	716	1676	1953	1858	1086	303	12	1
	CDH26.7	2999	0	0	6	22	189	688	811	798	411	73	2	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	1548	147	151	141	133	133	132	132	101	99	87	130	156
	PrecMax	2020	267	266	254	341	370	343	231	221	167	176	242	430
	PrecMin	930	56	66	21	38	19	35	46	4	16	0	78	64
	PrecStd	264	58	64	55	70	96	70	57	54	41	52	43	87
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	23.6	27.5	28.7	31.2	34.8	34.6	36.6	34.0	30.7	24.8	22.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	18.7	21.3	24.8	27.2	30.0	32.8	33.0	33.9	32.5	28.8	22.2	20.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	16.7	19.3	22.6	25.8	28.9	31.7	32.0	32.4	30.9	26.7	20.5	18.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	14.8	17.0	20.6	24.1	27.6	30.6	30.9	30.9	29.2	24.8	18.8	16.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(36) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	12.5	13.2	12.1	11.9	10.7	11.3	12.6	13.5	13.4	10.5
		MCDBR	11.0	13.2	14.1	14.1	13.5	13.4	11.9	13.1	14.0	15.4	14.3	10.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.315	0.321	0.359	0.391	0.430	0.458	0.500	0.510	0.421	0.366	0.334	0.314
	taud		2.511	2.490	2.409	2.309	2.243	2.187	2.100	2.062	2.303	2.440	2.472	2.529
	Ebn at Noon		896	928	914	895	860	832	795	778	839	864	863	875
	Edh at Noon		83	94	111	129	139	147	160	163	122	98	86	78
(44) All-Sky Solar Radiation	RadAvg		2.49	3.00	4.16	5.35	5.90	6.27	6.00	5.55	4.84	3.88	2.83	2.13
	RadStd		0.20	0.38	0.38	0.36	0.44	0.49	0.34	0.33	0.42	0.51	0.29	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+94

Nomenclature: See separate page